probability of all possible genotypes of offsprings in a genetic cross.The Law of Dominance and the Law of Segregation give suitable explanation to Mendel's monohybrid cross.

**Reciprocal cross** – In one experiment, the tall pea plants were pollinated with the pollens from a true-breeding dwarf plants, the result was all tall plants. When the parental types were reversed, the pollen from a tall plant was used to pollinate a dwarf pea plant which gave only tall plants. The result was the same - All tall plants. Tall (♀) x Dwarf (♂) and Tall (♂) x Dwarf (♀) matings are done in both ways which are called reciprocal crosses.The results of the reciprocal crosses are the same. So it was concluded that the trait is not sex dependent. The results of Mendel's monohybrid crosses were not sex dependent.

The gene for plant height has two alleles: Tall (T) x Dwarf (t). The phenotypic and genotypic analysis of the crosses has been shown by Checker board method or by Forkline method.

### 2.3.1 Mendel's analytical and empirical approach

Mendel chose two contrasting traits for each character. So it seemed logical that two distinct factors exist. In $F_1$ the recessive trait and its factors do not disappear and they are hidden or masked only to reappear in ¼ of the $F_2$ generation. He concluded that tall and dwarf alleles of $F_1$ heterozygote segregate randomly into gametes. Mendel got 3:1 ratio in $F_2$ between the dominant and recessive trait. He was the first scientist to use this type of quantitative analysis in a biological experiment. Mendel's data is concerned with the proportions of offspring.

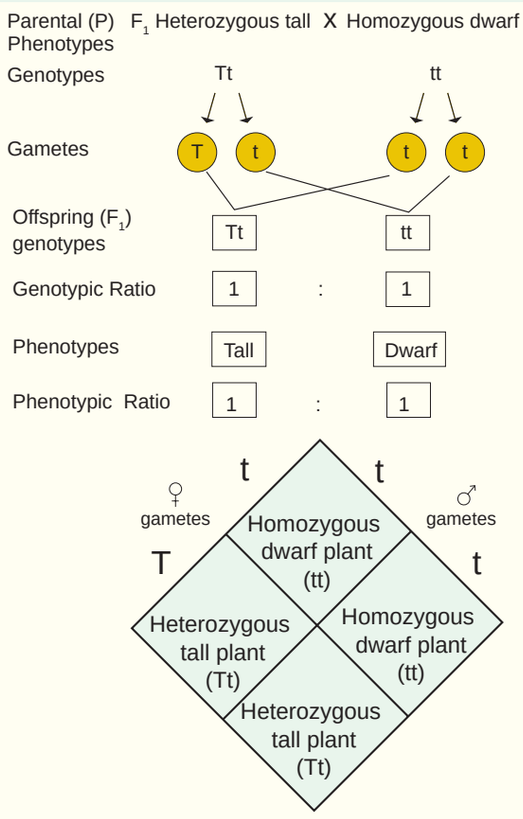
Mendel's analytical approach is truly an outstanding scientific achievement. His meticulous work and precisely executed breeding experiments proposed that discrete particulate units of heredity are present and they are transmitted from one generation to the other. Now they are called as genes. Mendel's experiments were well planned to determine the relationships which govern hereditary traits. This rationale is called an empirical approach. Laws that were arrived from an empirical approach is known as empirical laws.

### 2. 3.2 Test cross

Test cross is crossing an individual of unknown genotype with a homozygous recessive.

In Mendel's monohybrid cross all the plants are tall in $F_1$ generation. In $F_2$ tall and dwarf plants were in the ratio of 3:1.Mendel self pollinated dwarf $F_2$ plants and got dwarf plants in $F_3$ and $F_4$ generations. So he concluded that the genotype of dwarf was homozygous (tt). The genotypes of tall plants TT or Tt from $F_1$and $F_2$ cannot be predicted. But how we can tell if a tall plant is homozygous or heterozygous? To determine the genotype of a tall plant Mendel crossed the plants from $F_2$ with the homozygous recessive dwarf plant. This he called a test cross. The progenies of the test cross can be easily analysed to predict the genotype of the plant or the test organism. Thus in a typical test cross an organism (pea plants) showing dominant phenotype (whose genotype is to be determined) is crossed with the recessive parent instead of self crossing. Test cross is used to identify whether an individual is homozygous or heterozygous for dominant character.

**If heterozygous tall test cross**



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